PAPER

The Besserwisser Effect: A Complexity-Theoretic Proof that Criticism Cannot Substitute for Production

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Abstract

We provide a formal account of a widely observed empirical regularity: individuals with no production experience in a domain exhibit systematic overconfidence when evaluating artifacts produced by others. We call this the *Besserwisser Effect*. Assuming $\mathsf{P} \neq \mathsf{NP}$, we prove that the Production Problem—finding a quality artifact in an expressive domain—is NP-hard, while the Verification Problem—judging a given artifact—lies in P. This complexity gap is categorical: no polynomial accumulation of evaluation operations can solve an NP-hard production task (Theorem 1). An information-theoretic complement, which holds unconditionally, shows that k evaluation operations generate at most k bits of novel domain knowledge, while a single production operation generates $\Omega(\log |D|)$ bits, where $|D|$ is the domain size (Proposition 1). We apply the framework to educational systems, showing that standard curricula are structurally evaluation-training machines, and derive three falsifiable empirical predictions.

Keywords: computational complexity, P vs NP, verification, production, expertise, Dunning-Kruger, information theory, education

1 Introduction

A reliable empirical observation crosses domains: people who have never created in a field nonetheless judge its products with high confidence. Film critics who have never directed, economic commentators who have never built a model, code reviewers who have never shipped a product—all exhibit what we call the *Besserwisser Effect*. The gap has two components: critics are overconfident relative to their actual accuracy, and their accuracy is lower than that of practitioners with comparable exposure to existing artifacts.

Prior explanations appeal to cognitive bias [3] or incentive structures: critics bear no cost of error. Both are correct but incomplete. They explain the *persistence*

of the gap but not its *origin*. We propose a more fundamental account.

The key insight is that *finding* a quality solution and *checking* a proposed solution are problems of different computational complexity. For the broad class of expressive production tasks—creative work, software, economic models, scientific arguments—verification is polynomial while production is NP-hard. Under $\mathsf{P} \neq \mathsf{NP}$ [1, 2], this gap is not gradual but categorical: no finite accumulation of evaluation experience closes it.

A complementary information-theoretic argument, which requires no complexity-theoretic assumptions, shows that evaluation generates at most one bit per operation while production generates $\log_2 |D|$ bits. For any realistic domain the gap is exponential and grows with domain expressiveness.

The paper is organized as follows. Section 2 defines the formal objects. Section 3 proves the main theorem. Section 4 derives the information bound. Section 5 develops a dynamical characterization. Section 6 applies the result to educational systems. Section 7 states falsifiable predictions. Section 8 states limitations.

2 Definitions

Definition 1 (Domain and quality function)**.** *Let D be a finite domain of artifacts with $|D| = 2^{\Omega(n)}$. A quality function $Q : D \to [0, 1]$ is* efficiently evaluable *(computable in* $\mathrm{poly}(n)$ *time given an artifact) but NP-hard to maximize.*

The efficiently-evaluable assumption is natural: a reader can assess whether a paper is good faster than they can write one.

Definition 2 (Production Problem, Prod)**.** *Given $(D, Q, \tau \in [0, 1])$: find $a \in D$ with $Q(a) \geq \tau$, or output $\perp$.*

Definition 3 (Verification Problem, Ver)**.** *Given (D, a, τ): decide whether $Q(a) \geq \tau$.*

Definition 4 (Besserwisser strategy)**.** *A Besserwisser strategy $\mathcal{B}$ is a* $\mathrm{poly}(n)$*-time algorithm that calls* Ver *on externally supplied $a_1, \ldots, a_k \in D$ and outputs judgments $j_i \in \{0, 1\}$. $\mathcal{B}$ never constructs a new element of D.*

3 The Besserwisser Theorem

Lemma 1 (Prod is NP-hard)**.** *For Q NP-hard to optimize,* Prod *is NP-hard.*

Proof. Reduce 3-SAT to Prod. Given formula φ on n variables, set $D = \{0, 1\}^n$, $Q(a) = 1$ iff $a \models \varphi$ and $Q(a) = 0$ otherwise, $\tau = 1$. Then φ is satisfiable iff $\textsc{Prod}(D, Q, 1) \neq \perp$. Since 3-SAT is NP-complete [1], Prod is NP-hard. □

Lemma 2 (Ver is in P)**.** *Compute $Q(a)$ in* $\mathrm{poly}(n)$ *and return $Q(a) \geq \tau$.*

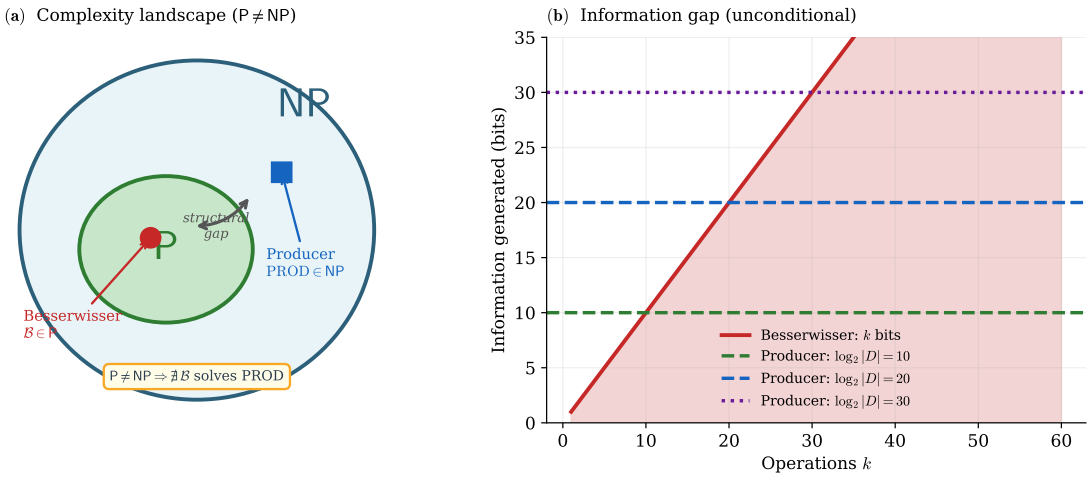

Figure 1: **(a)** Complexity landscape under $\mathsf{P} \neq \mathsf{NP}$. Besserwisser strategies are confined to P (green); production lies in $\mathsf{NP} \setminus \mathsf{P}$ (blue). The structural gap cannot be closed by accumulating evaluations. **(b)** Information generated per operation. Each evaluation yields one bit; one production yields $\log_2 |D|$ bits. For $|D| = 2^{20}$ (a modest domain), a single production operation equals 20,000 evaluations informationally.

Theorem 3 (Besserwisser Theorem)**.** *Assume* $\mathsf{P} \neq \mathsf{NP}$*. No Besserwisser strategy* $\mathcal{B}$ *solves* PROD*, regardless of how many evaluations it performs.*

Proof. Suppose $\mathcal{B}$ solves PROD. Since $\mathcal{B}$ runs in poly(n) time and PROD is NP-hard, we obtain $\mathsf{P} = \mathsf{NP}$. Contradiction. □

Corollary 4 (Accumulation is irrelevant)**.** *Performing* k *additional evaluations does not reduce the complexity of* PROD *for any* $k \in \mathbb{N}$*.*

Proof. The complexity class of PROD is independent of polynomial preprocessing. $\mathsf{P} \neq \mathsf{NP}$ implies PROD $\notin \mathsf{P}$ regardless of prior observations. □

Corollary 4 makes a precise, testable claim: watching more film reviews does not make one a better filmmaker; reading more economic commentary does not make one a better economist. The class boundary is not a gradient—it is categorical.

4 Information-Theoretic Bound

Proposition 5 (Information gap)**.** *Let* $|D| = 2^n$*.*

(i) k *evaluations generate at most* k *bits of domain knowledge.*

(ii) One production operation generates at least $n = \log_2 |D|$ *bits (under a uniform prior over* D*).*

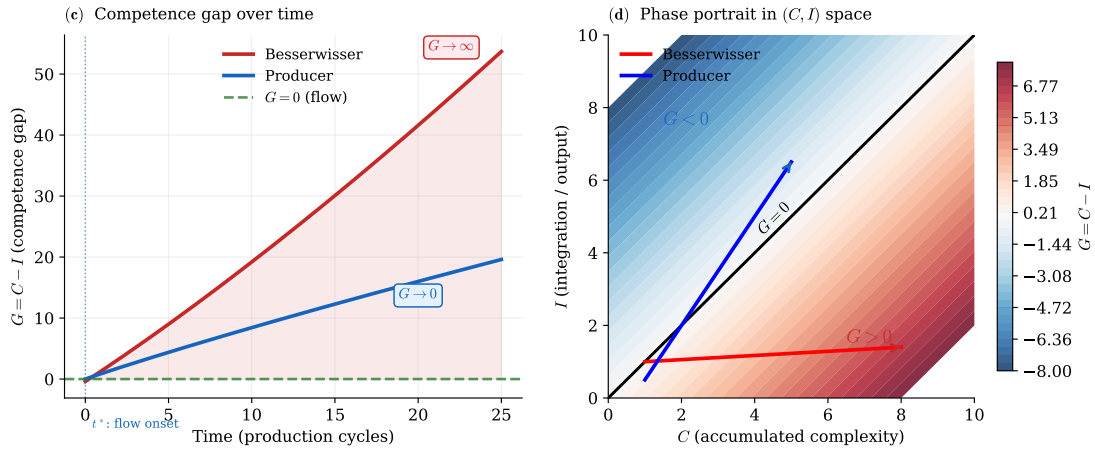

Figure 2: **(c)** Competence gap $G = C - I$ over time. Besserwisser diverges ($G \to \infty$); Producer converges toward $G = 0$ after t^* production cycles. **(d)** Phase portrait in (C, I) space. Red trajectory: Besserwisser accumulates C with near-zero I. Blue trajectory: Producer crosses $\{G = 0\}$ into flow.

For $k = \text{poly}(n) \ll 2^n$, the gap is exponential.

Proof. (i) Each VER call returns one bit. Shannon's source coding theorem bounds k calls at k bits [4].

(ii) Selecting $a^* \in D$ under a uniform prior over D carries entropy $H = \log_2 |D| = n$ bits. For $|D| = 2^{20}$ and $k = 100$: production yields 20 bits per operation; evaluation yields 1 bit. The ratio is $20 : 1$ per operation, and $2^{20}/100 \approx 10{,}000 : 1$ cumulatively. □

This result requires no complexity-theoretic assumption—it is a direct consequence of Shannon's theorem and holds unconditionally. It provides the stronger empirical claim: even if $\mathsf{P} = \mathsf{NP}$ were somehow shown, the information asymmetry would persist.

5 Dynamical Characterization

Let $C(t)$ denote accumulated domain knowledge (content consumed, opinions formed) and $I(t)$ denote integrated domain knowledge (artifacts produced, problems solved). Define the *competence gap*:

$$G(t) \;=\; C(t) - I(t) \tag{1}$$

Critic dynamics. Each evaluation increases C by α and I by $\varepsilon \ll \alpha$:

$$G_{\mathcal{B}}(k) = G_0 + (\alpha - \varepsilon)k \;\longrightarrow\; \infty \tag{2}$$

The critic's competence gap diverges monotonically.

Producer dynamics. Production drives I superlinearly:

$$G_{\mathcal{P}}(t) = \beta t - \gamma t^{1+\delta}, \quad \gamma, \delta > 0 \tag{3}$$

with zero-crossing at $t^* = (\beta/\gamma)^{1/\delta}$, after which the producer enters the flow regime ($G < 0$). Figure 2(d) shows both trajectories. No sequence of horizontal moves (evaluation) reaches the line $\{G = 0\}$—the geometric restatement of Theorem 3.

6 Application: Educational Systems

Standard educational curricula are structurally evaluation-training machines. Their primary instruments are verification tasks: examinations (given a question, evaluate the answer), essays (evaluate an argument), presentations (evaluate an exposition). The production of genuinely novel artifacts is rare and, where it occurs, typically extracurricular.

A student completing 12 years of school and 5 years of university accumulates approximately 17 years of intensive VER training with negligible PROD experience. By Theorem 3 and Corollary 4, this produces agents optimized for evaluation with no formal production capacity. The Dunning-Kruger phenomenon is not a pathology of this system—it is its optimal output given the objective function being maximized (VER-score).

This yields a testable prediction: students with high extracurricular production activity will outperform high-grade students on prospective quality-prediction tasks a decade after graduation. Longitudinal panel data (e.g., SOEP) could directly test this.

7 Falsifiable Predictions

Prediction 1 (Calibration gap). Critics with zero production experience will show higher confidence but lower prospective accuracy than producers with equal artifact exposure:

$$\mathbb{E}[\text{Confidence} \mid \mathcal{B}] > \mathbb{E}[\text{Confidence} \mid \mathcal{P}], \quad \mathbb{E}[\text{Accuracy} \mid \mathcal{B}] < \mathbb{E}[\text{Accuracy} \mid \mathcal{P}] \tag{4}$$

Data: Blind evaluation experiment; critics vs. producers judge new artifacts in creative, technical, or analytic domains.

Prediction 2 (Competence gap divergence). The ratio $C(t)/I(t)$ diverges for pure evaluators and stabilizes for producers over longitudinal observation.

Data: Longitudinal records tracking consumption vs. production per individual over ≥ 3 years.

Prediction 3 (Education-production correlation). Hobby intensity during student years correlates positively with artifacts-produced a decade later; GPA correlates near zero or negatively after controlling for hobby intensity.

Data: SOEP panel data or alumni survey linking student-era hobby intensity to career-era production output.

Falsification. If critics with zero production experience outperform producers in prospective quality prediction after controlling for domain exposure, the empirical application fails. The complexity-theoretic proof remains valid but its applied interpretation requires revision.

8 Limitations

Domain scope. NP-hardness of PROD requires Q to be hard to optimize. For domains with linear, efficiently invertible quality functions the gap closes. The theorem applies to expressive domains (natural language, software, economics, creative work) where quality is not efficiently invertible.

Hybrid actors. Most practitioners combine production and evaluation. The theory predicts a continuous relationship between production volume and calibration quality, not a binary critic-or-producer classification.

Empirical validation. Predictions 1–3 require individual-level longitudinal data at scale. No large-sample test has been conducted yet.

The $\mathsf{P} \neq \mathsf{NP}$ assumption. The main theorem relies on the leading conjecture in theoretical computer science, unrefuted for 50 years. The information-theoretic Proposition 5 holds unconditionally.

9 Conclusion

We have proved the Besserwisser Theorem: under $\mathsf{P} \neq \mathsf{NP}$, no strategy confined to evaluation can solve the Production Problem, regardless of how many evaluations it performs. The information-theoretic bound holds unconditionally: k evaluations yield $\leq k$ bits of domain knowledge; one production operation yields $\log_2 |D|$ bits. Applied to educational systems, the framework predicts that standard curricula produce well-calibrated evaluators and structurally underprepared producers. Three falsifiable predictions are stated with concrete data sources. The conclusion is direct: there is no polynomial path from criticism to production competence. The only route is production.

Code and data. https://github.com/hakvinv/besserwisser-effect.

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